CHECKER - Automated SOC Attack Simulation and Logging

Project Overview

CHECKER is an automated SOC attack simulation and logging tool.
The script discovers active hosts on the local network, allows the user to choose a target, executes controlled security checks, and records every selected attack in /var/log/checker.log.
At the end of the session, the script displays a final report that includes the detected IP addresses, executed attacks, detailed attack results, and the log file location.

Operational Logic Flow

1. Root Permission Check:
   The script checks whether it is running with root permissions. Root access is required because the log file is saved under /var/log/checker.log.
2. Required Tools Check:
   The script verifies that Nmap, Hydra, Hping3, and Arpspoof are installed. If a required tool is missing, the user can choose whether to install it.
3. Log File Preparation:
   The script creates or opens /var/log/checker.log and adds a new session title with the current date and time.
4. Network Range Detection:
   The script detects the local IP address and automatically creates a /24 network range.
5. IP Address Discovery:
   Nmap is used to discover available IP addresses on the local network. The detected addresses are saved into an array and displayed to the user.
6. Target Selection:
   The user can choose a specific target IP address or select a random target from the detected hosts.
7. Attack Selection:
   The user can choose one of the five attacks or select a random attack.
8. Multiple Attack Execution:

   After each attack, the user can choose whether to run another attack in the same session.
9. Final Report:
   At the end, the script displays the detected IP addresses, executed attacks, detailed results, and the log file location.

Attack Types and Execution Logic

The project includes five controlled attacks: TCP Port Scan, UDP Scan, SSH Weak Credentials, SYN Flood, and ARP Spoofing.
These attacks represent four different attack classes: Reconnaissance, Credential Attack, Denial of Service, and Man in the Middle.
Each attack is implemented as a separate function and executed against the selected target IP address.

Attack Type	Execution Logic	Result Handling
TCP Port Scan	Uses Nmap to scan the selected target for the top 200 common TCP ports. The -Pn option allows the scan to continue even if the target does not respond to ping.	The script extracts open TCP ports and displays them in the final report.
UDP Scan	Uses Nmap UDP scan to check the top 20 common UDP ports on the selected target.	The script extracts open UDP results and displays a clean summary. The full UDP output is saved to the log file.
SSH Weak Credentials	Uses Nmap to verify that SSH port 22 is open. If SSH is available, Hydra tests selected usernames against a built-in weak password list.	The script shows each tested username with a FOUND or NOT FOUND status. The full Hydra output is saved to the log file.
SYN Flood	Uses Hping3 to send a controlled burst of 1000 TCP SYN packets to TCP port 80 on the selected target.	The script records the target, attack class, target port, number of packets sent, and execution result.
ARP Spoofing	Uses Arpspoof to perform a controlled Man in the Middle simulation between the selected target and the default gateway.	The script records the target, attack class, gateway IP, network interface, and execution result.

Requirements Implementation

At least three attacks
The script includes five attacks: TCP Port Scan, UDP Scan, SSH Weak Credentials, SYN Flood, and ARP Spoofing. These attacks cover four different attack classes: Reconnaissance, Credential Attack, Denial of Service, and Man in the Middle.

Separate function for each attack:
Each attack is implemented in its own function: attack_port_scan, attack_udp_scan, attack_ssh_weak_credentials, attack_syn_flood, and attack_arp_spoof.

Clear attack descriptions
The attack menu displays the name and purpose of each attack before selection.

Display available IP addresses
The script scans the local network and displays all detected IP addresses in a numbered list.

Manual or random attack selection
The user can choose a specific attack or select the Random Attack option.

Invalid input handling

The script validates user input. If the selected value is invalid, an error message is displayed and the program terminates.

Logging under /var/log

Every selected attack is recorded in /var/log/checker.log.

Log details

The log includes the attack type, execution time, target IP address, and full technical output.

Functions and comments

The script is divided into clear functions and includes comments explaining the main actions.

Tools Used in the Attack Functions

The project uses four main security tools across the five attack functions: Nmap, Hydra, Hping3, and Arpspoof.
Nmap is used for host discovery, TCP and UDP port scanning, and SSH port validation. Hydra is used in the SSH Weak Credentials function to test the collected usernames against the built-in weak password list. Hping3 is used to perform the controlled SYN Flood simulation. Arpspoof is used to perform the controlled Man in the Middle simulation.

Attack Function	Attack Type	Tool Used	Explanation
attack_port_scan	TCP Port Scan	Nmap	Scans the selected target for the top 200 common TCP ports.
attack_udp_scan	UDP Scan	Nmap	Scans the selected target for the top 20 common UDP ports.
attack_ssh_weak_credentials	SSH Weak Credentials	Nmap & Hydra	Nmap verifies that SSH port 22 is open. Hydra then tests the collected usernames against the built-in weak password list.
attack_syn_flood	SYN Flood	Hping3	Sends a controlled burst of 1000 TCP SYN packets to the selected target.
attack_arp_spoof	ARP Spoofing	Arpspoof	Performs a controlled Man in the Middle simulation between the selected target and the gateway.

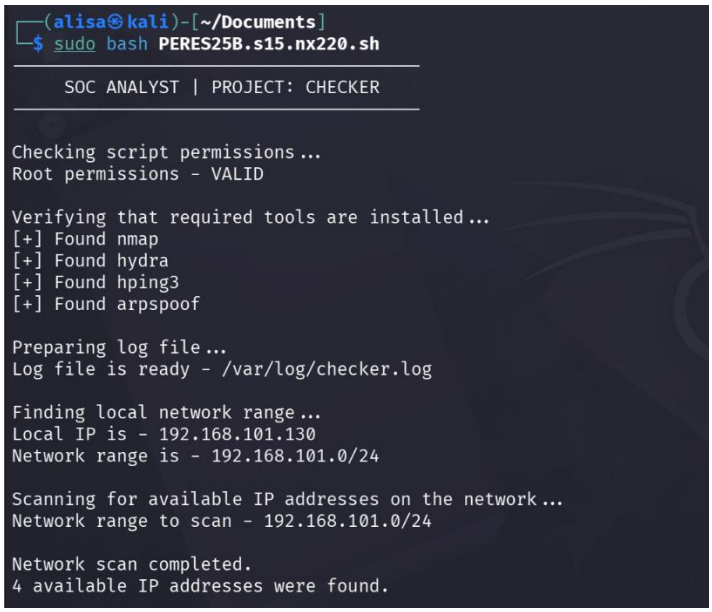
Conclusion

The script discovers available network targets, allows manual or random attack selection, executes five controlled attacks across four different attack classes, validates user input, records each selected activity into /var/log/checker.log, and displays a final session report.

Output Screenshots

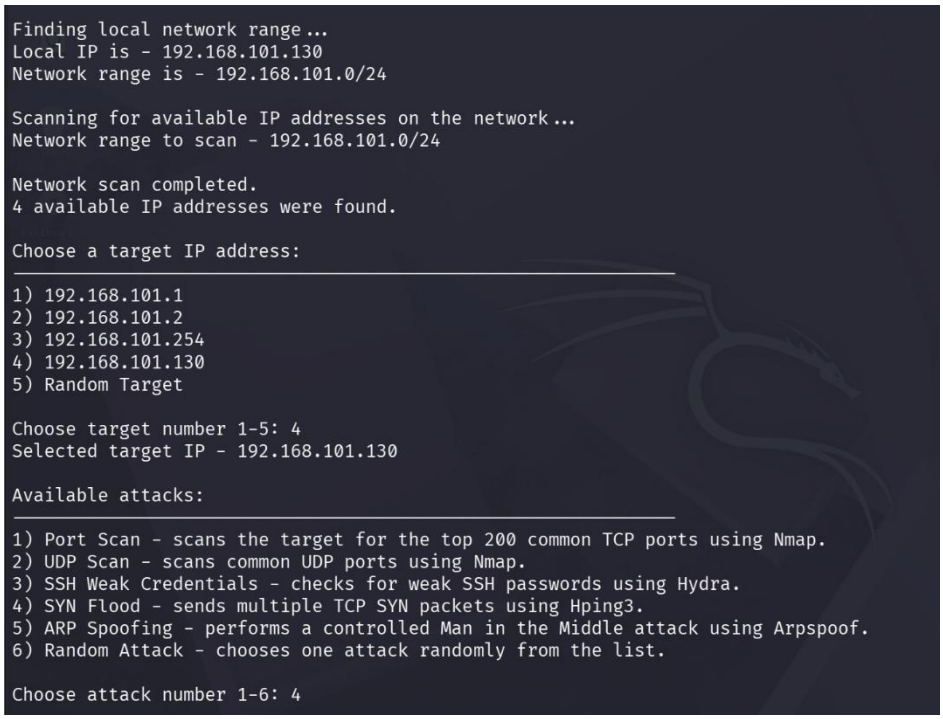
Startup and Network Discovery

This screenshot shows the beginning of the script execution. The program validates that it is running with root permissions, checks that the required tools are installed, prepares the log file, detects the local IP address, creates the network range, and scans the network for available IP addresses.



Target and Attack Selection

This screenshot shows the user selection stage. The script displays all detected target IP addresses and allows the user to choose either a specific target or a random target. It also displays the available attack types and allows the user to choose a specific attack or a random attack.



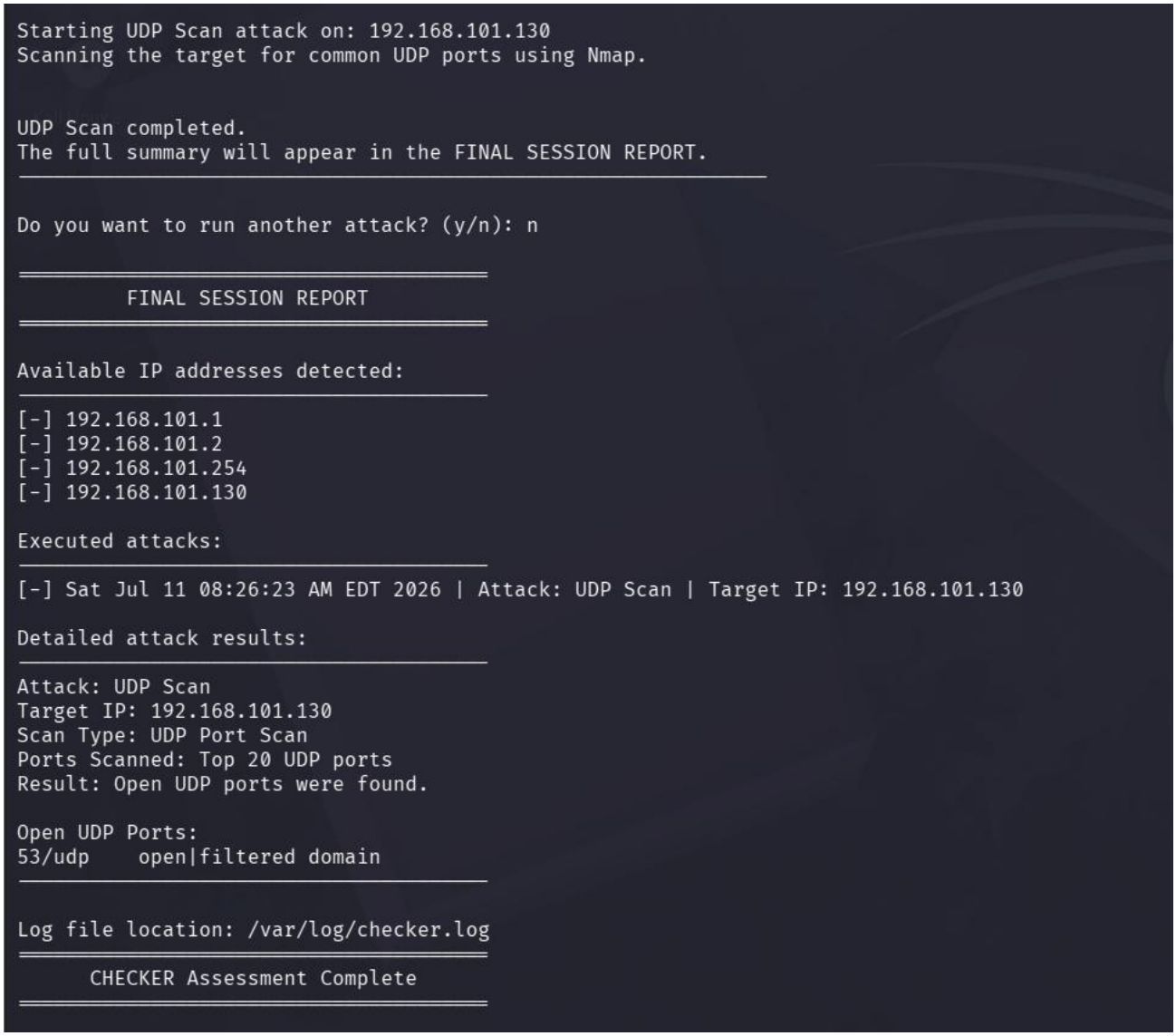
TCP Port Scan Result

This screenshot shows the execution of the TCP Port Scan attack. The selected target is scanned for the top 200 common TCP ports using Nmap. The final report displays the open TCP ports that were found, including their port numbers, states, and detected services.



UDP Scan Result

This screenshot shows the execution of the UDP Scan attack. The selected target is scanned for common UDP ports using Nmap. The final report displays the UDP scan result and saves the complete technical output into the log file.



SSH Weak Credentials Result

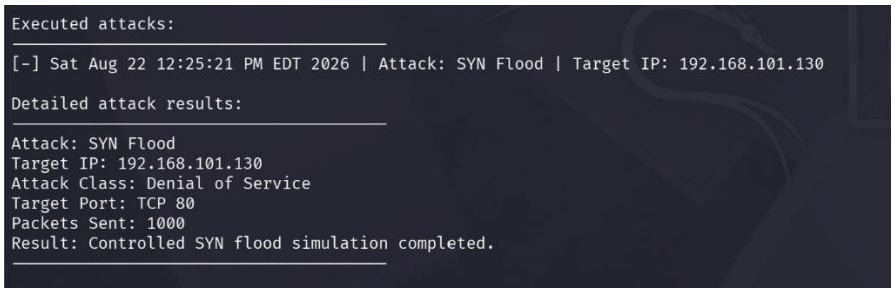
This screenshot shows the execution of the SSH Weak Credentials attack. The script first verifies that SSH port 22 is open, collects the usernames to be tested, and then uses Hydra to test them against the built-in weak password list.

The final report clearly shows which usernames were tested and whether weak credentials were found.



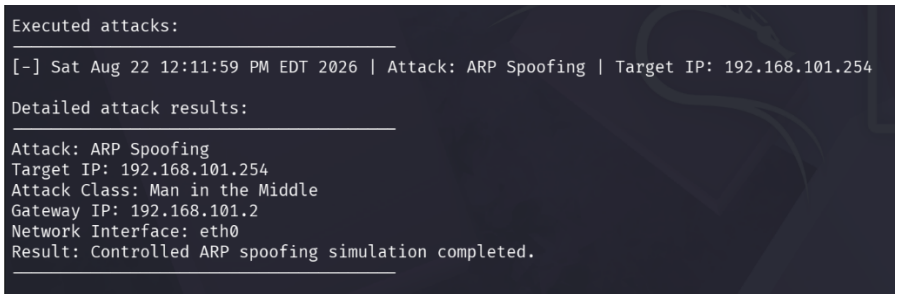
SYN Flood Result

This screenshot shows the execution of the SYN Flood attack. The script sends a controlled burst of TCP SYN packets to the selected target using Hping3. The final report confirms that 1000 packets were sent and the Denial of Service simulation completed successfully.



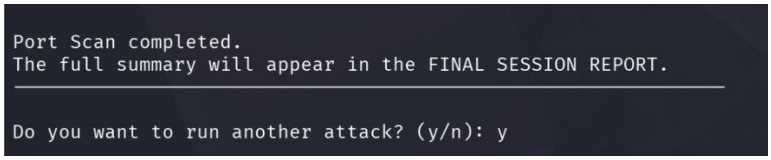
ARP Spoofing Result

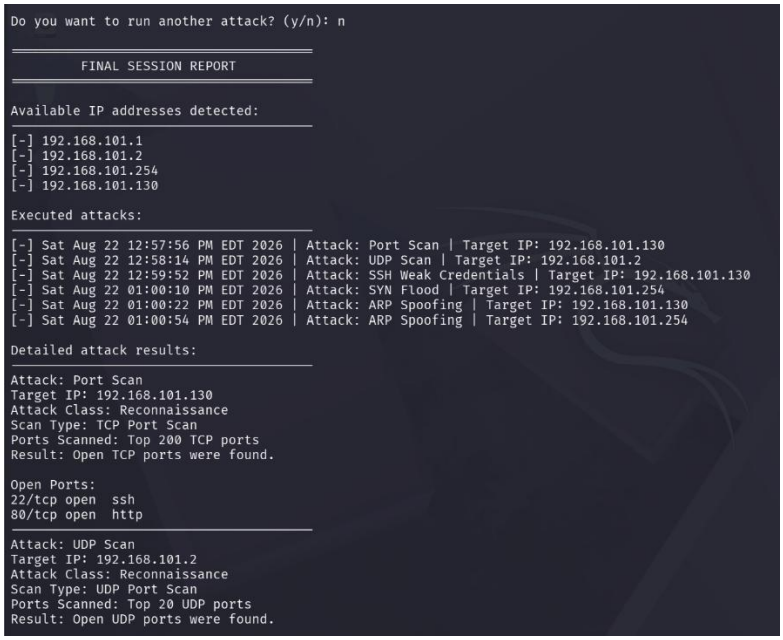
This screenshot shows the execution of the ARP Spoofing attack. The script performs a controlled Man in the Middle simulation against the selected target, using the detected network interface and default gateway. The final report confirms that the ARP spoofing simulation completed successfully.

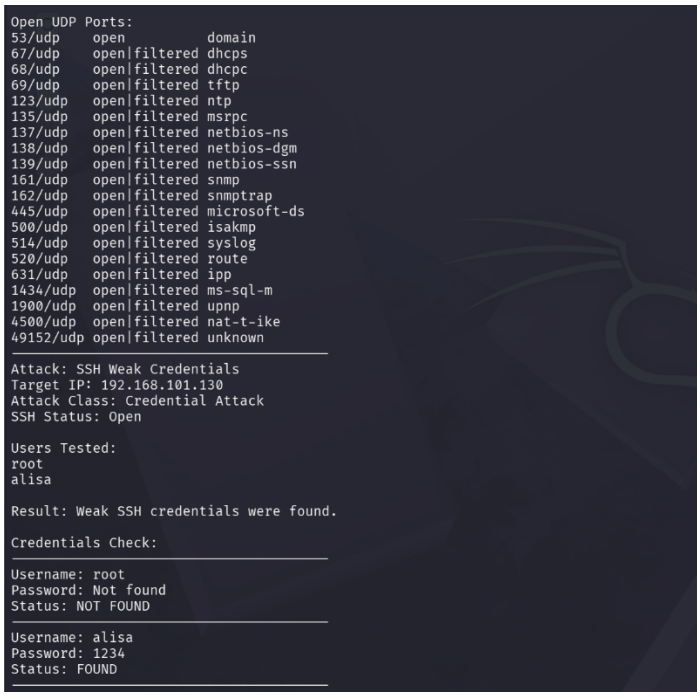


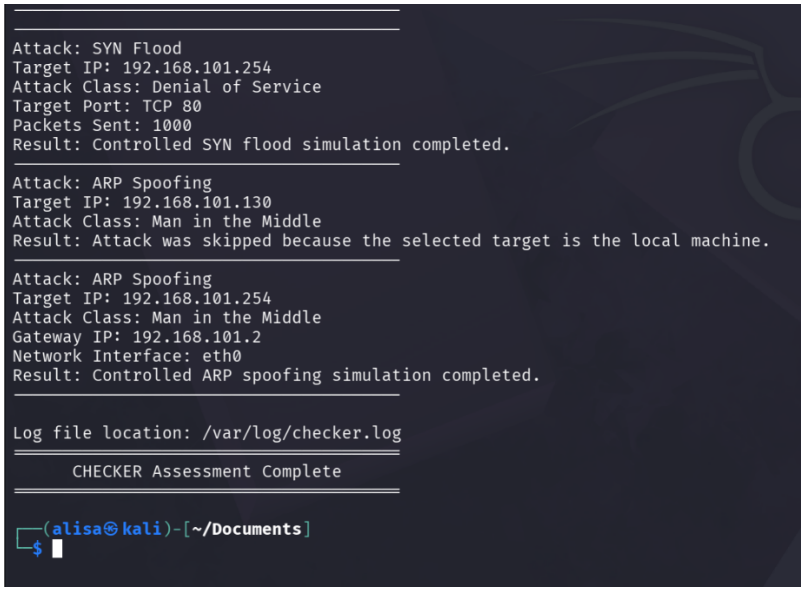
Multiple Attacks in One Session

This screenshot demonstrates that the script can execute multiple attacks during the same session. After each attack, the user can choose to run another attack without restarting the program. The final report summarizes all executed attacks and their detailed results together.



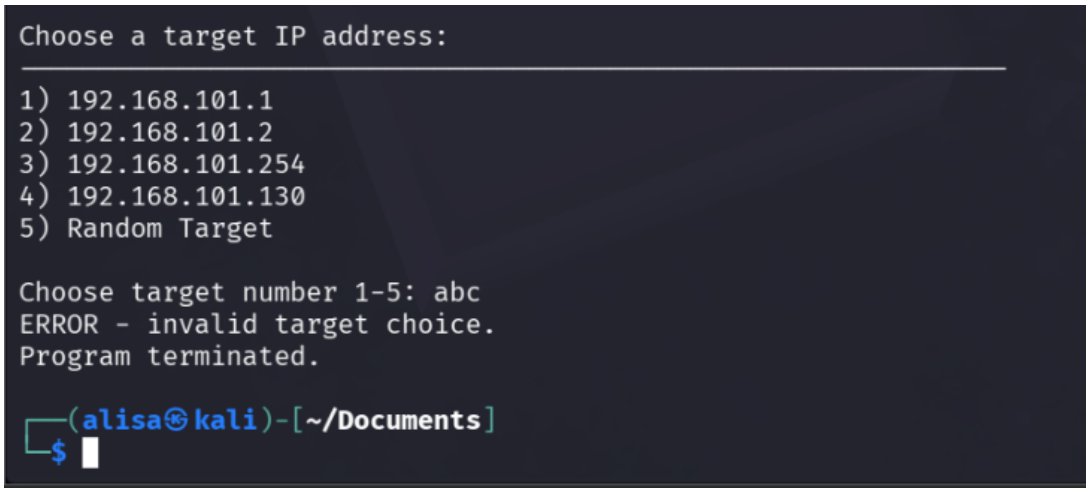






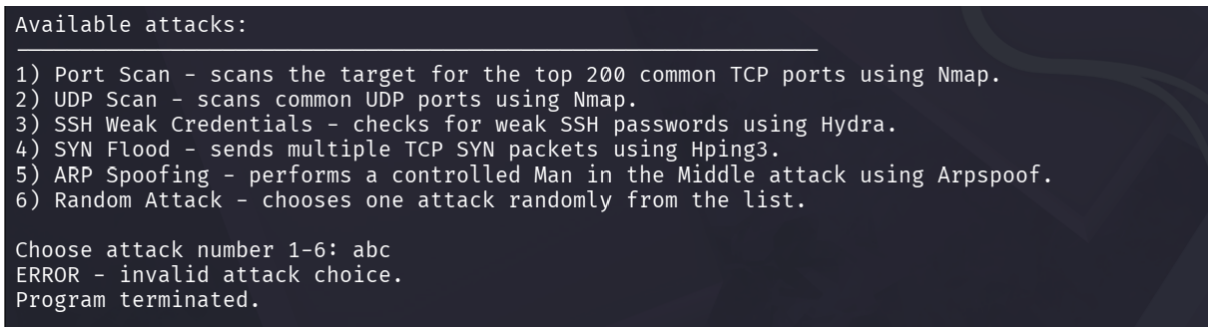
Invalid Target Input

This screenshot shows the script's input validation for target selection. When the user enters an invalid target value, the program displays an error message and terminates safely.



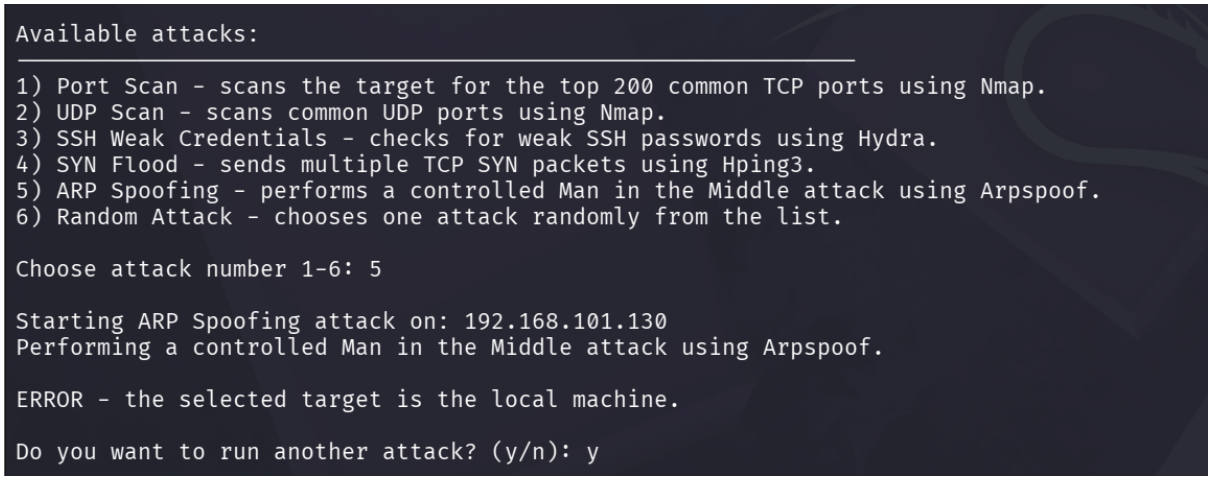
Invalid Attack Input

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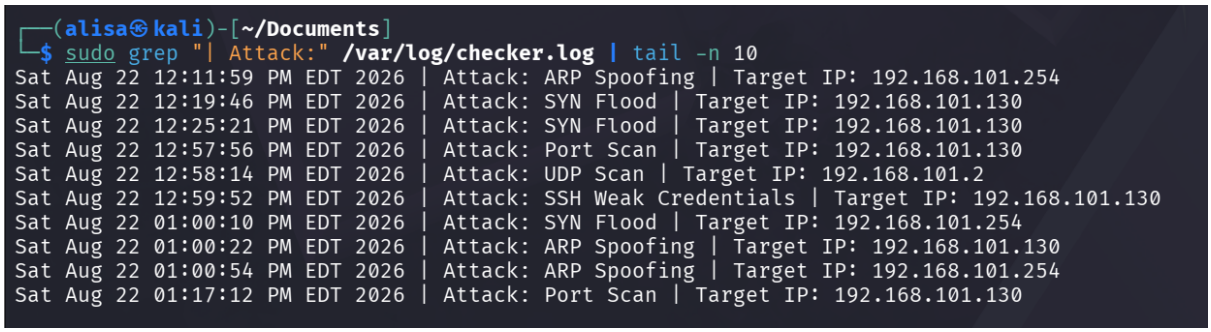
ARP Spoofing Local Target Validation

This screenshot shows the script's target validation for the ARP Spoofing attack. When the selected target is the local machine, the script detects it and stops the attack. A clear error message is displayed to prevent the local machine from being used as the ARP Spoofing target.



Log Summary

This screenshot shows the CHECKER log entries for the executed attacks. The log records the attack type, target IP address, and execution time.



Run Without Sudo / Root Error

The screenshot proves that the script validates execution permissions before starting the main process.
If the user runs the script without sudo, the program stops and displays a clear error message, preventing execution without the required permissions.

